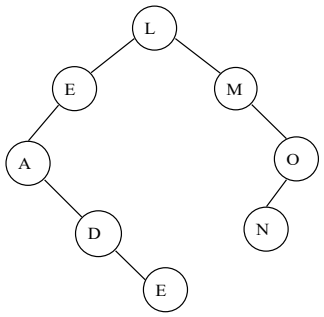
Data Structures Solutions – Senior Division

1. **00-01 C4 Data Structures**

Starting at the bottom, the diagram shows that the depth of the E is 4, 16
the D and N have depths of 3, the A and O have depths of 2 and the E



and M have depths of 1. The sum the depths is 16

2. **00-01 C4 Data Structures**

In a stack the letters are popped on a last in first out basis. L
The table gives the order of the letters popped:

1	2	3	4	5	6	7	8	9	10	11	12
A	N	S	M	0	R	E	G	A	S	E	D

The letters left in the stack are REVL so L would be the next letter popped.

3. **01-02 C4 Data Structures**

When Grumpy is compared to Dopey, go right. When compared to 25
Sleepy, go left. When compared to Happy go left and insert at node.
Eight nodes can be attached. They have a path length of 25 = 5(3)+ 2 + 2(4).

4. **01-02 C4 Data Structures**

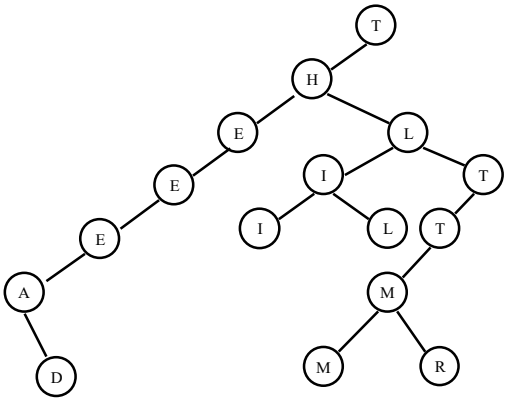
The following is the result of the operations on the stack: I

B, BI, BIN, BI, BID, BI, BIS, and BI.

The next item to be popped is I.

5. **02-03 C4 Data Structures**

The binary search tree is as follows: 4



There are 4 nodes with exactly 2 children.

6. **02-03 C4 Data Structures**

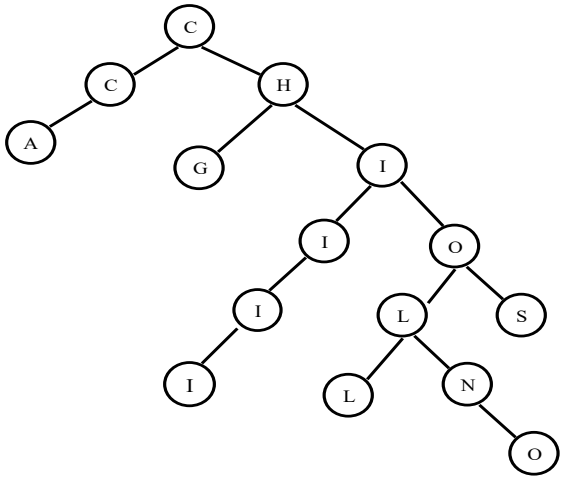
The queue is built as follows: I

MAR, ARY, RYPO, POP, OPPI, PPINS, INS

The next letter to be POP'ed is the I.

7. **03-04 C4 Data Structures**

The binary search tree is as follows: 47



The internal path length is 2 * 1 + 3 * 2 + 2 * 3 + 3 * 4 + 3 * 5 + 1 * 6 =

2+6+6+12+15+6 = 47.

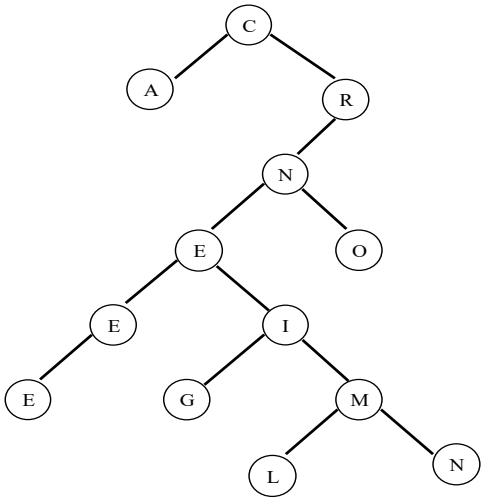
8. **03-04 C4 Data Structures**

A postorder transversal visits the left and right subtrees then visits the nodes. As shown

Therefore, it is A C G I I I L O N L S O I H C

9. **04-05 C4 Data Structures**

The sum of the depths at each level is: 2+2+6+8+15+12 = 45. 45



10. **04-05 C4 Data Structures**

The queue uses the principal of FIFO. Y

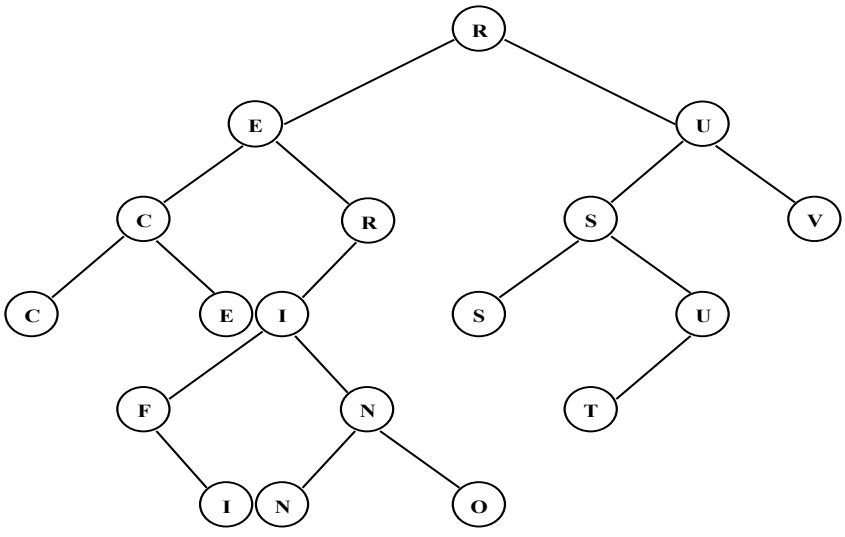
The contents of the queue changes as follows:

H, HE, E, ER, ERS, RS, S, SH, SHE, HE, E, EMPTY, Y

11. **05-06 C4 Data Structures**

The external path length is calculated as follows: 88

6 * 6 + 3 * 5 + 7 * 4 + 3 * 3 = 36 + 15 + 28 + 9 = 88



12. **05-06 C4 Data Structures**

The queue is constructed using FIFO and changes as follows: G

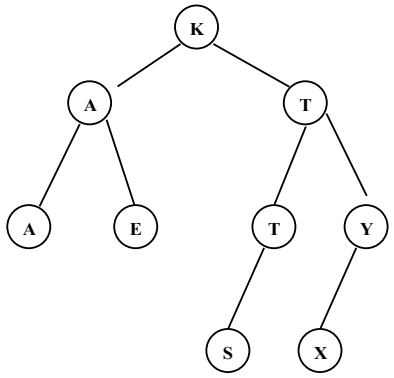
ABC, CBA, BA, BAD, DAB, AB, B, BEF, EF, FE, E, EG, GE.

The next item removed would be G.

13. **06-07 C4 Data Structures**

The binary search tree is shown below. 16

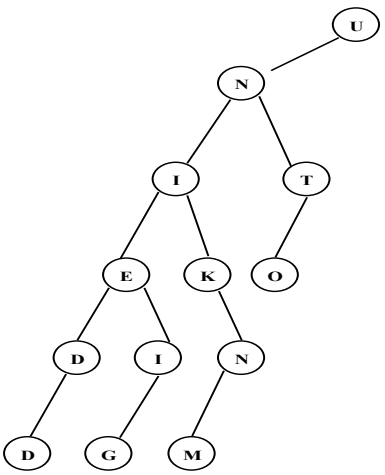
The path length is found by adding 0 + 1 + 1 + 2 + 2 + 2 + 2 + 3 + 3 = 16.

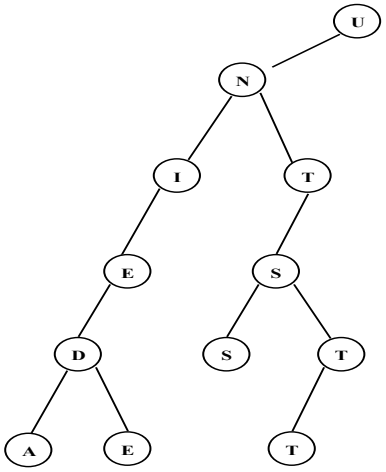


14. **06-07 C4 Data Structures**

The first tree has 6 nodes with one child and the 1

second tree has 5 nodes with one child.

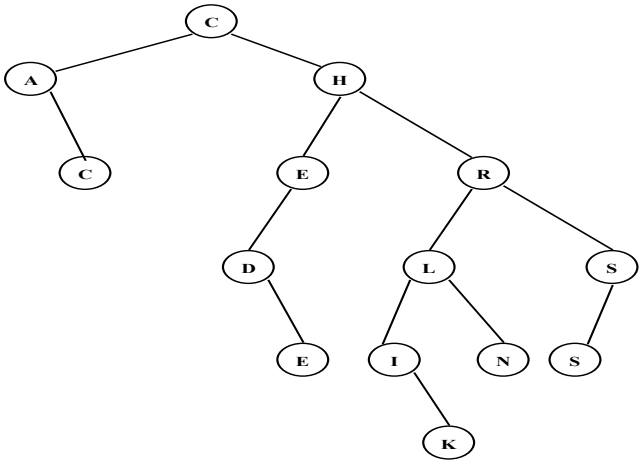




15. **07-08 C4 Data Structures**

The binary search tree is shown below. 28

The path length is found by adding 0 + 2*1 + 3*2 + 3*3 + 4*4 + 5 = 38.



16. **07-08 C4 Data Structures**

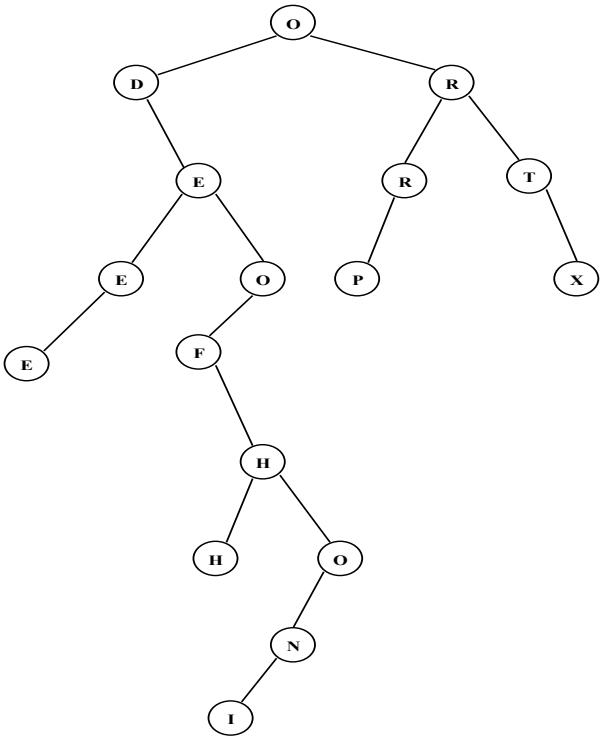
The stack is constructed as follows: S

BLE, BL, LB, LBAK, KABL, KA, AK, AKH, A, AO, OA, O, OSE, ESO, ES.

The next popped item is S since stacks are LIFO.

17. **08-09 C4 Data Structures**

There are 8 nodes with only\one child: D,E,O,F,O,N,R,T 8



18. **08-09 C4 Data Structures**

The construction is as follows: N

ANI, AN, ANM, ANMA, MA, MAT, MA, MAI, AI, AIO, IO, ION

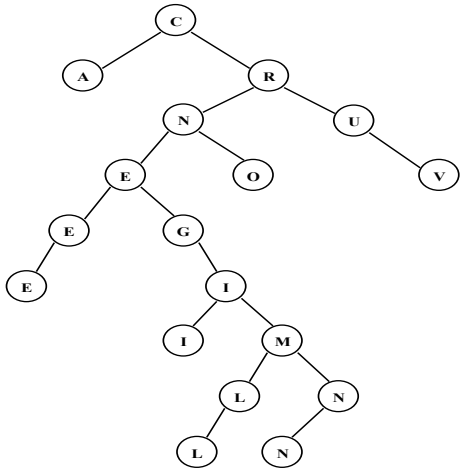
The next popped item is N since it ended as a stack.

19. **09-10 C4 Data Structures**

The binary search tree is shown on the right. 75

The path length is found by adding

1*0 + 2*1 + 2*2 + 3*3 + 4*2 + 5*2 + 6*2 + 7*2 + 8*2 = 75



20. **09-10 C4 Data Structures**

The stack is constructed as follows: C

A, AB, ABC, ABCD, ABCED, ABCE, ABCEF, ABCE, AGBCE, AGBC

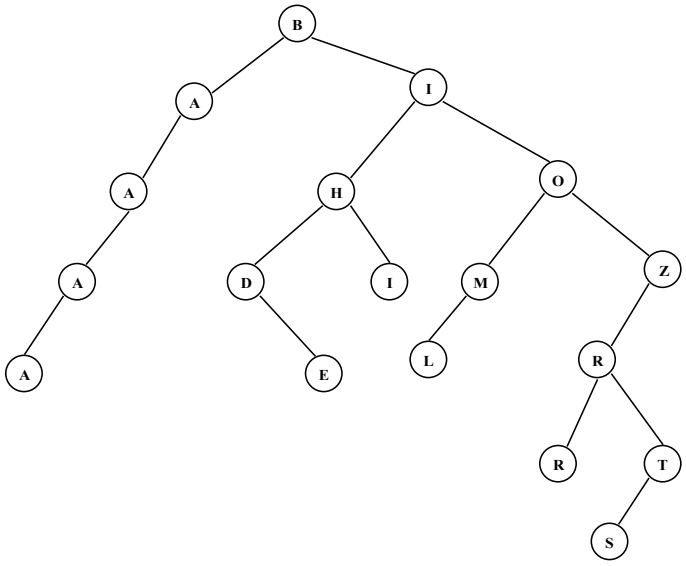
The next popped item is C since stacks are LIFO.

21. **10-11 C4 Data Structures**

The binary tree formed is shown on the right. 55

The path length is found by adding

1* 0 + 2*1 + 3*2 + 5*3 + 4*4 + 2*5 + 1*6 = 55



22. **10-11 C4 Data Structures**

The construction is as follows: S

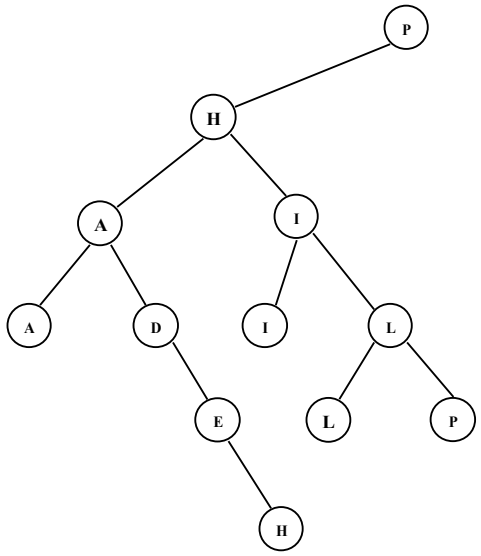
MU, UM, U, USE, ESU, SU, SUU, UUS, UUSM, USM, USMS

The next popped item is S since it ended as a stack.

23. **11-12 C4 Data Structures**

The internal path length is found by adding: 34

0 + 1*1+2*2+4*3+3*4+1*5 =34



24. **11-12 C4 Data Structures**

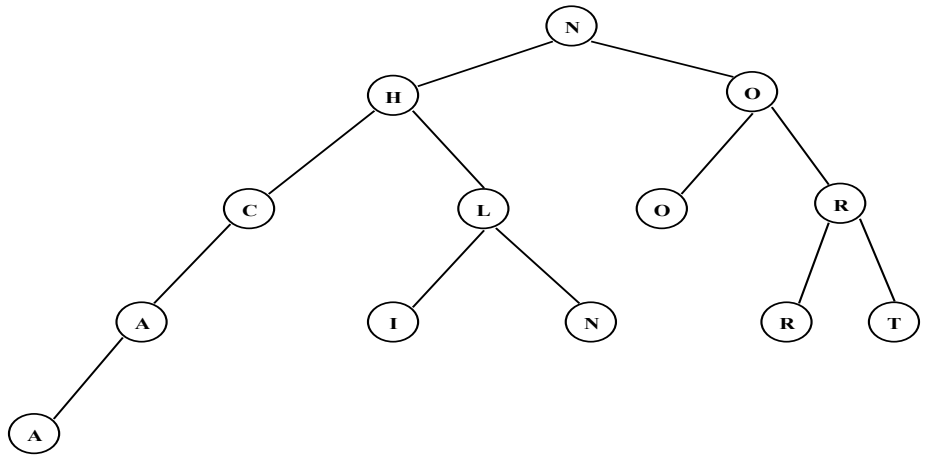
The construction is as follows: H, HE, HER, HERS, ERS, ERSH, RSH, A

SH, SHE, HE, HEY, HEYP, EYP, YP, YPA, YPAR, PAR, PARK, ARK.

The next popped item is A.

25. **12-13 C4 Data Structures**

The internal path length is: 2*1+4*2+5*3+1*4 = 29. 29

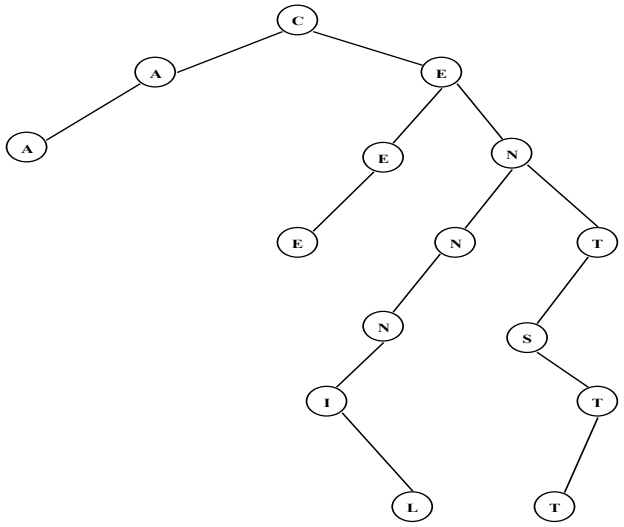


26. **12-13 C4 Data Structures**

The construction is as follows: M, ME, MED, ED, EDI, EDIT, EDITE, A
EDIT, DIT, DITR, DITRA, ITRA, TRA, TR, TRE, TREA, TREAN,
REAN, EAN, EANS, EANSE, ANSEA, NSEA
The next popped item is A since it ended as a stack.

27. **13-14 C4 Data Structures**

The external path length is: 0*1+1*2+3*3+4*4+2*5+2*6+4*7 = 77 77

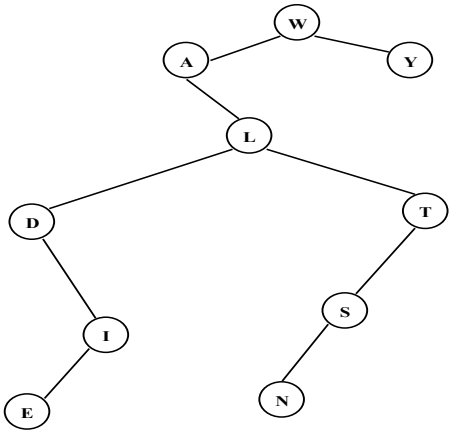


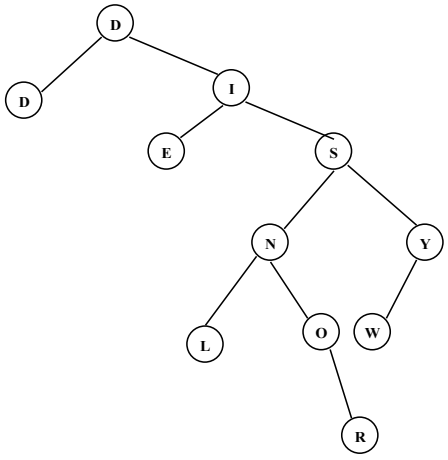
28. **13-14 C4 Data Structures**

The construction is as follows: B, BL, BLA, LA, LAC, LACK, O
ACK, CK, CKC, KC, C, CA, CAN, AN, ANY, NY, Y, YO, YON, ON.
The next popped item is O since it is a queue.

29. **14-15 C4 Data Structures**

WALTDISNEY DISNEYWORLD 1





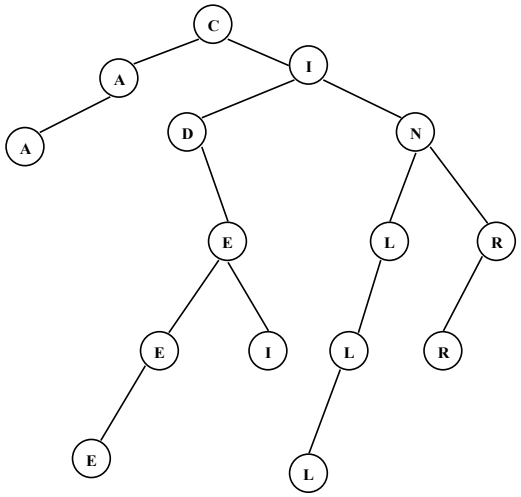
Internal path length
= 2*1 + 1*2 + 2*3 + 2*4 + 2*5 = 28

Internal path length
= 2*1 + 2*2 + 2*3 + 3*4 + 1*5 = 29

30. **14-15 C4 Data Structures**

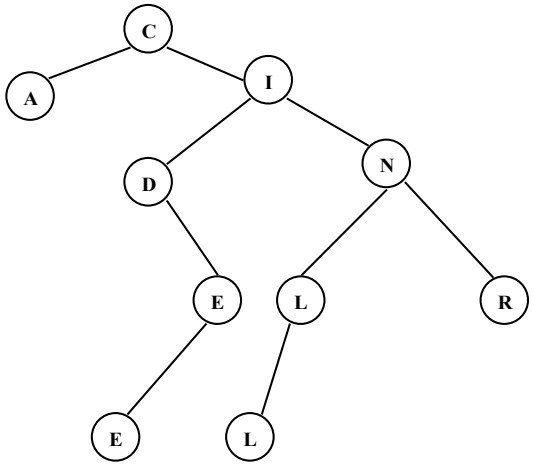
CINDERELLA

DEPTH = 5



CINDERELLAARIEL

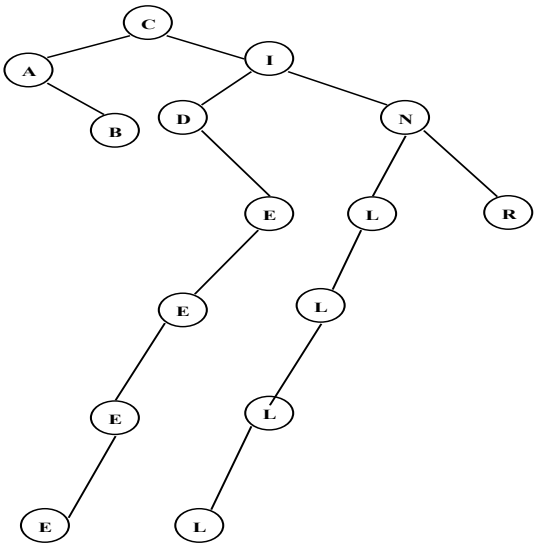
DEPTH = 4



BELLE

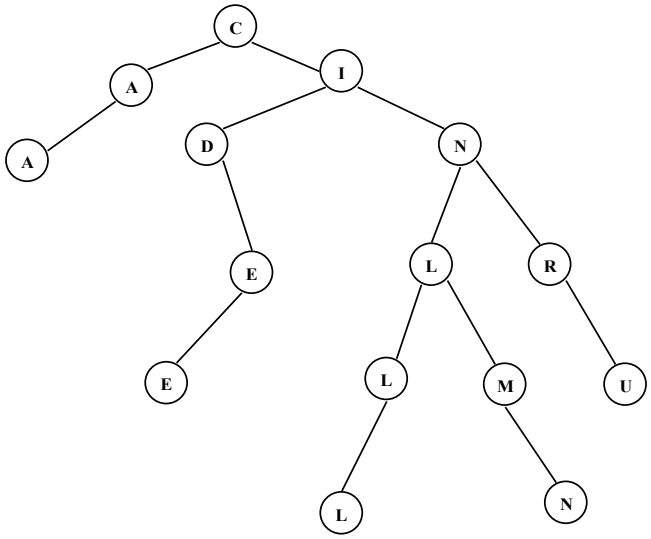
CINDERELLAMULAN

DEPTH = 6



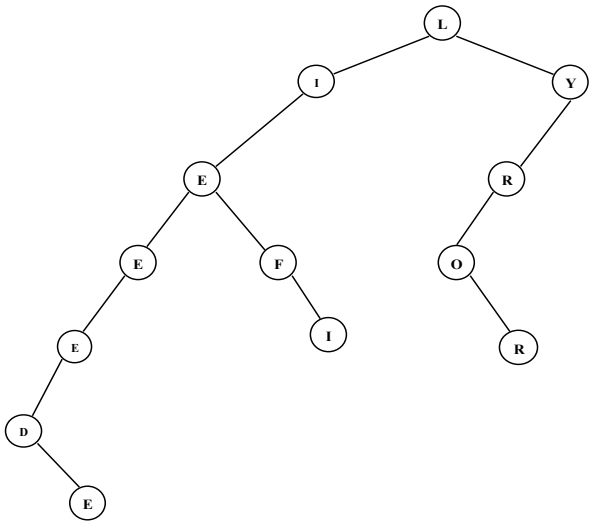
CINDERELLABELLE

DEPTH = 5



31. **15-16 C4 Data Structures**

The internal path length is: 2*1+2*2+3*3+3*4+1*5+1*6 = 38. 38



32. **15-16 C4 Data Structures**

The construction is as follows: (QUEUE) D, DA, A, O

(STACK) AR, ART, ARTM, ART, AR, (QUEUE) R, RO, ROU, OU,

(STACK) O, OT, (QUEUE) OTH,

The next POP'ed item is an O since it is a queue.